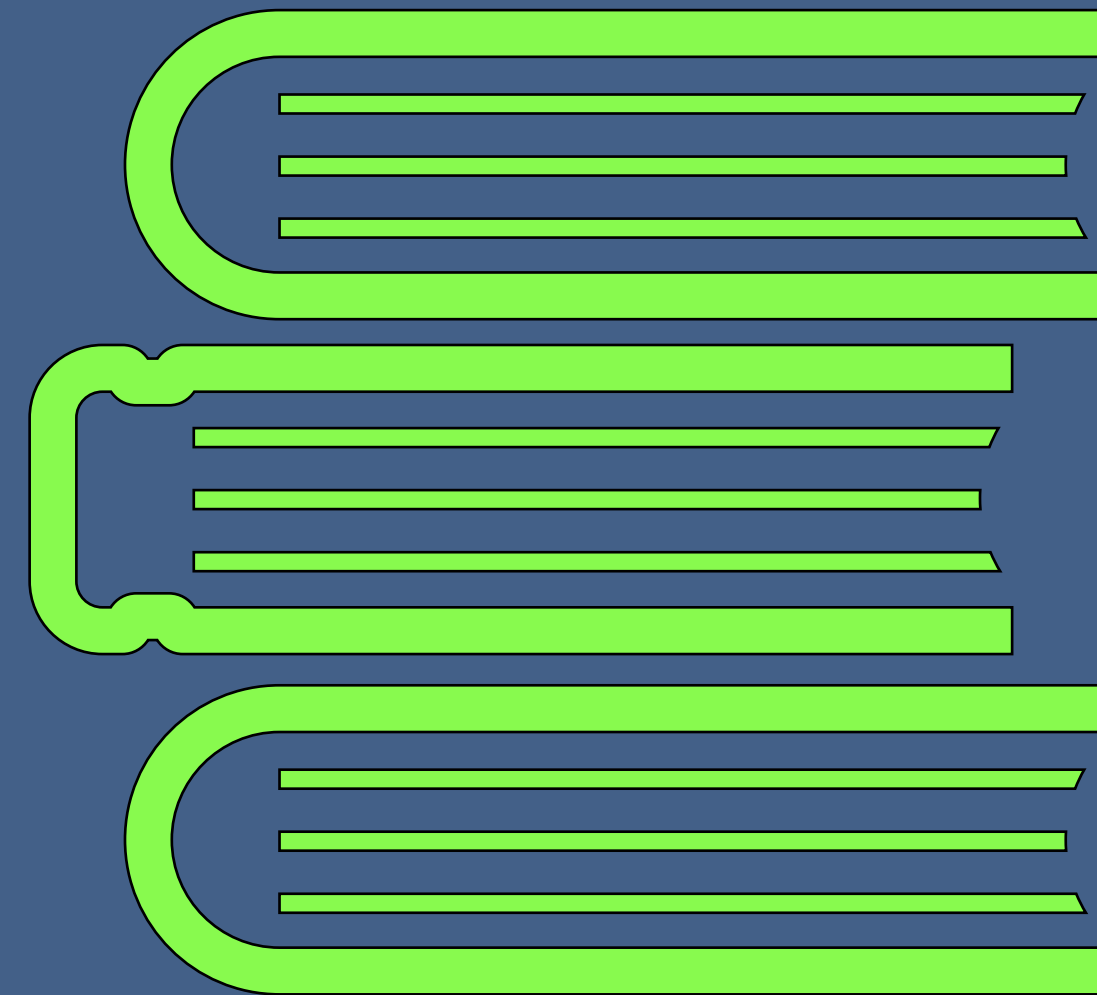
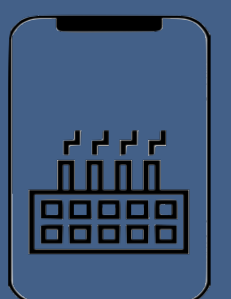


Stack



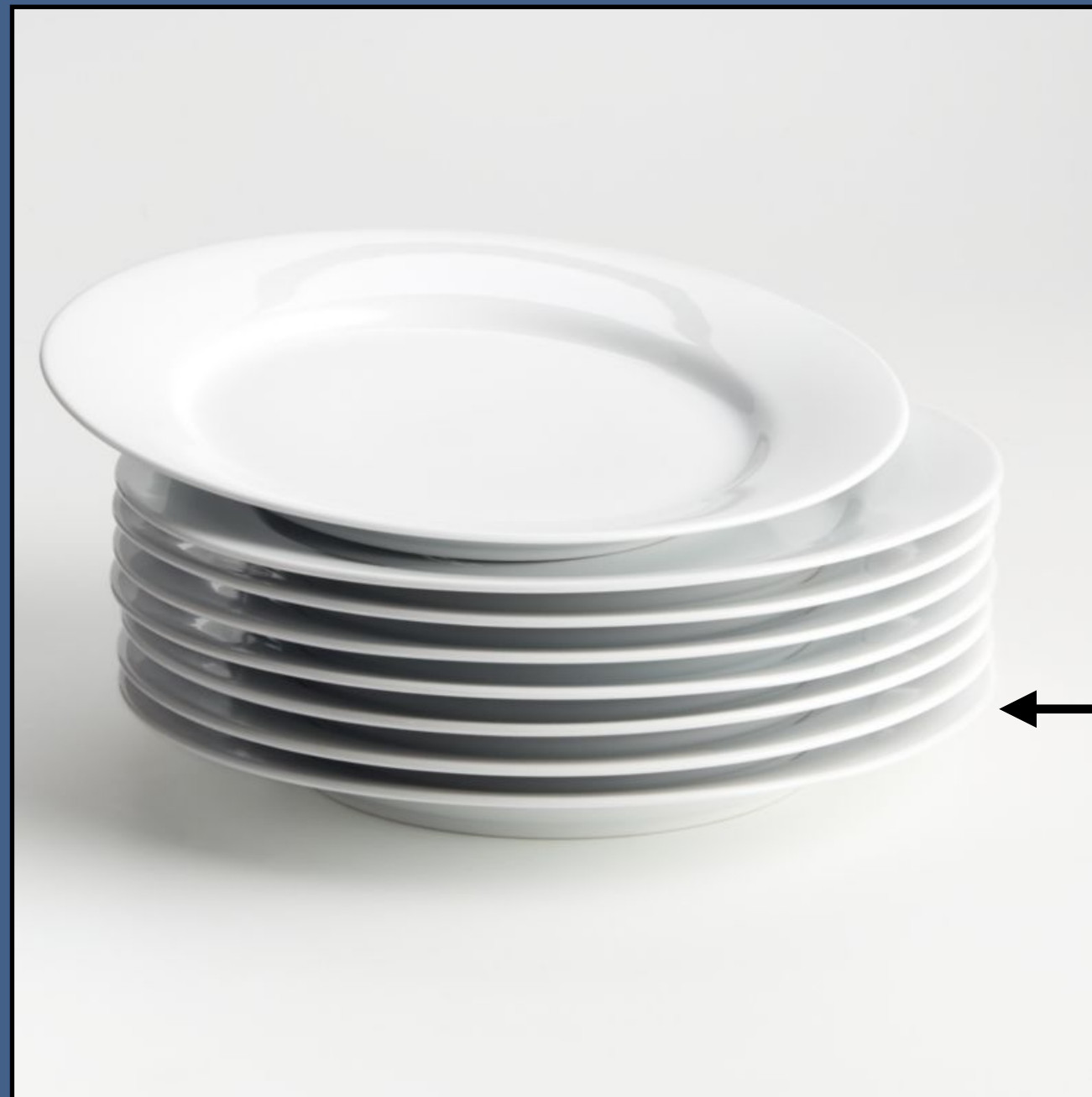
What is a Stack?

Stack is a data structure that stores items in a Last-In/First-Out manner.

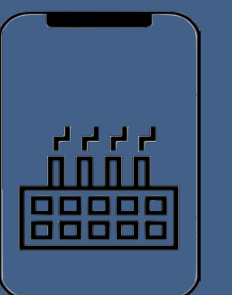


What is a Stack?

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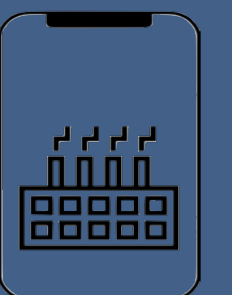
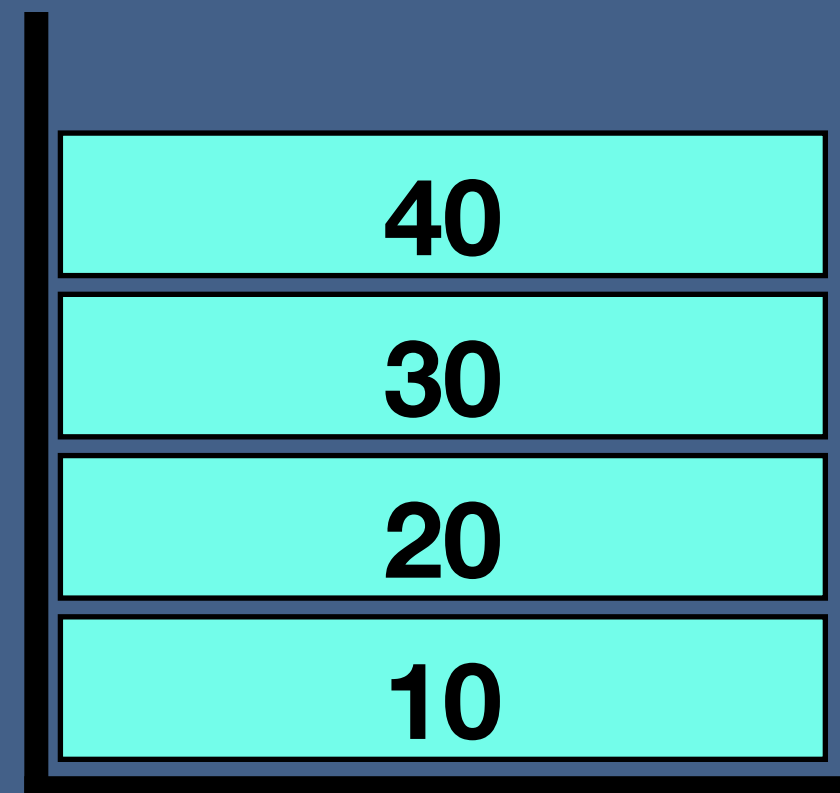


LIFO method

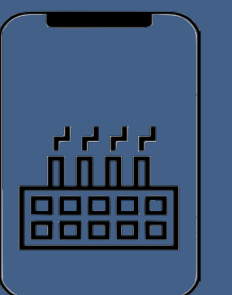
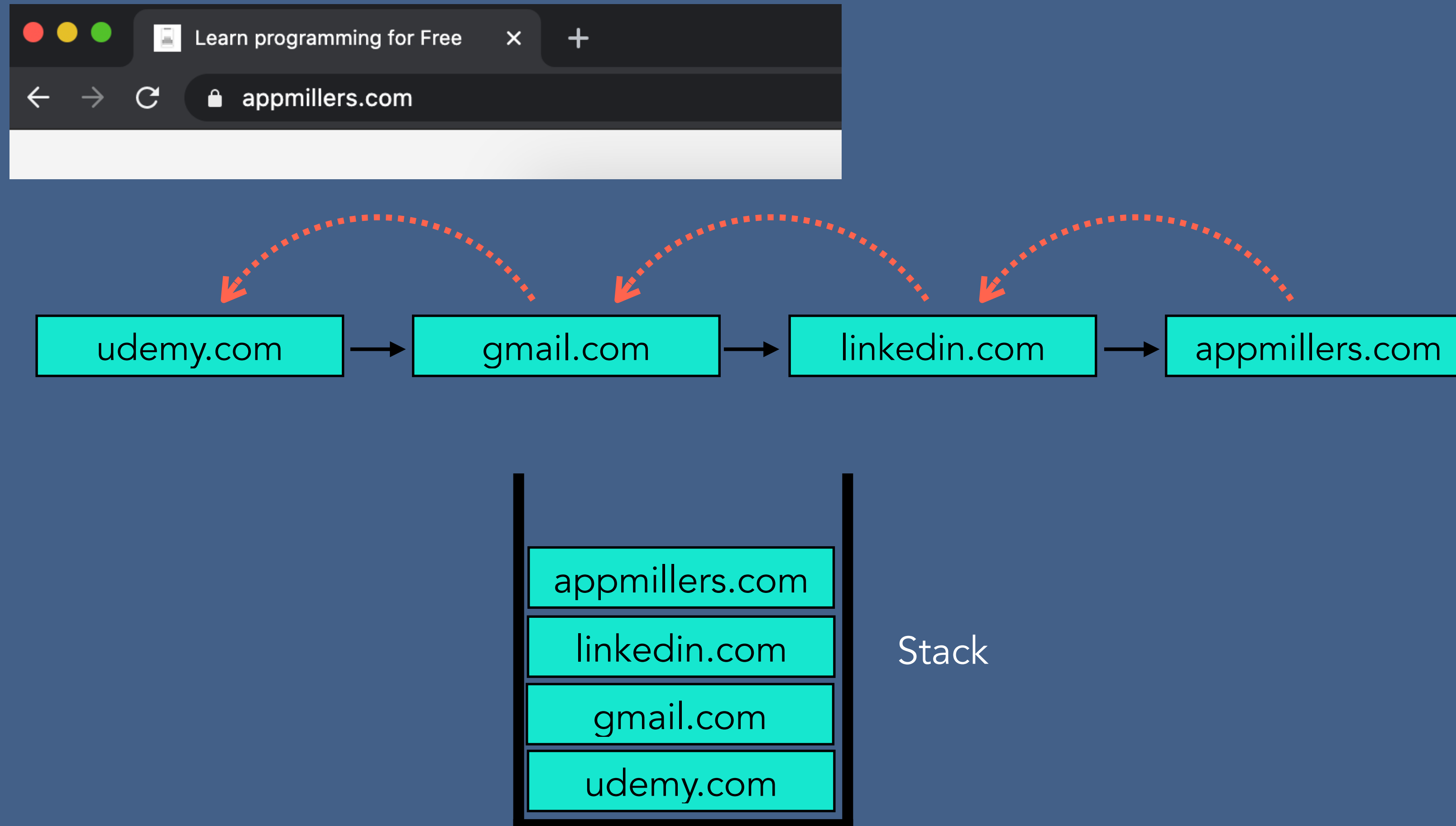


What is a Stack?

Stack is a data structure that stores items in a Last-In/First-Out manner.



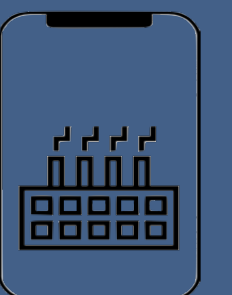
Why do we need Stack?



Stack Operations

- Create Stack
- Push
- Pop
- Peek
- isEmpty
- isFull
- deleteStack

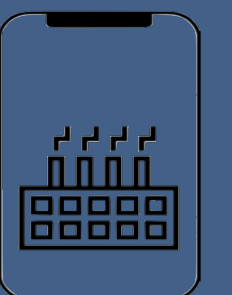
customStack()



Push Method

```
customStack = [ ]
```

```
customStack.push(1)
```

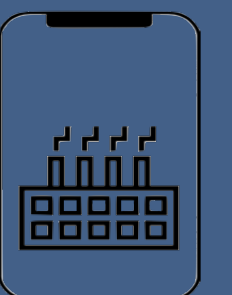



Push Method

```
customStack = [1]
```

```
customStack.push(2)
```

				1					

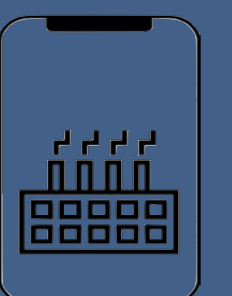


Push Method

```
customStack = [1,2]
```

```
customStack.push(2)
```

				2					
				1					

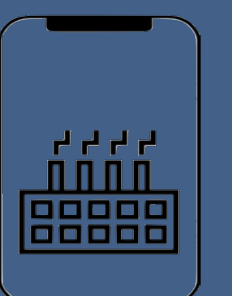


Push Method

```
customStack = [1,2,3]
```

```
customStack.push(4)
```

				3					
				2					
				1					

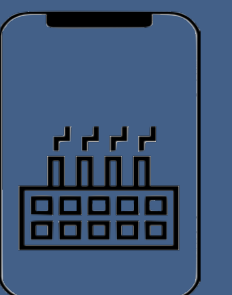


Push Method

```
customStack = [1,2,3,4]
```

```
customStack.push(4)
```

				4					
				3					
				2					
				1					

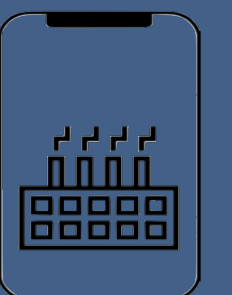


Pop Method

customStack = [1,2,3]4]

customStack.pop() → 4

				4					
				3					
				2					
				1					

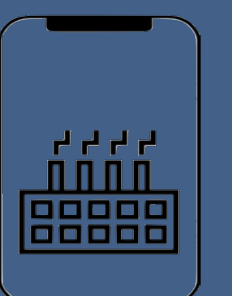


Pop Method

customStack = [1,2]3

customStack.pop() → 3

				3					
				2					
				1					

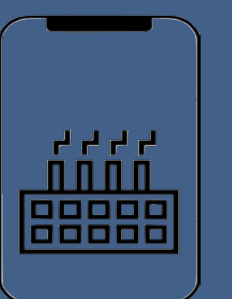


Pop Method

customStack = [1]2]

customStack.pop() → 2

				2					
				1					

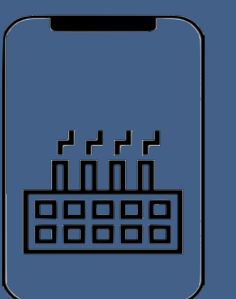


Pop Method

```
customStack = [1]
```

```
customStack.pop() → 1
```

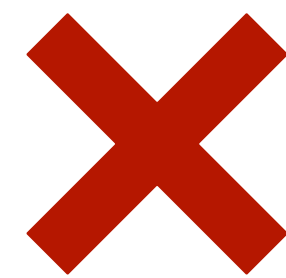
				1					



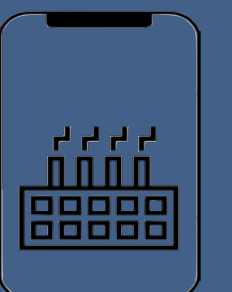
Pop Method

```
customStack = []
```

```
customStack.pop()
```



The stack is Empty

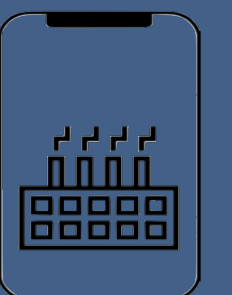


Peek Method

customStack = [1,2,3,4]

customStack.peek() $\longrightarrow$ 4

				4					
				3					
				2					
				1					

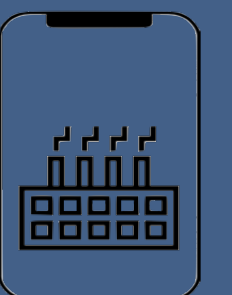


isEmpty Method

```
customStack = [1,2,3,4]
```

```
customStack.isEmpty() → False
```

				4					
				3					
				2					
				1					

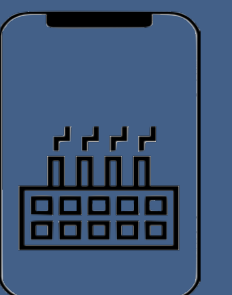


isFull Method

```
customStack = [1,2,3,4]
```

```
customStack.isFull() → False
```

				4					
				3					
				2					
				1					

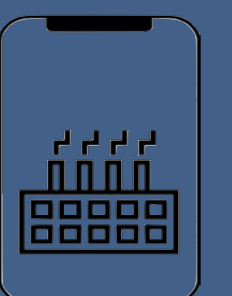


Delete Method

```
customStack = [1,2,3,4]
```

```
customStack.delete()
```

				4					
				3					
				2					
				1					



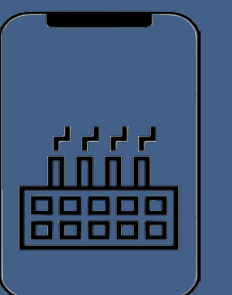
Stack Creation - Array vs Linked List

Stack using Array

- Easy to implement
- Fixed size

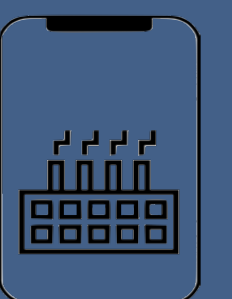
Stack using Linked List

- Variable size
- Implementation is not easy



Time and Space Complexity of Stack using Array

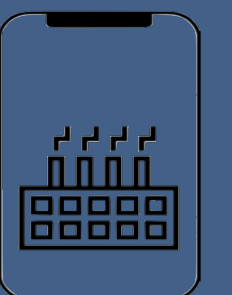
Stack	Time complexity	Space complexity
Create Stack	$O(1)$	$O(n)$
Push	$O(1)$	$O(1)$
Pop	$O(1)$	$O(1)$
Peek	$O(1)$	$O(1)$
isEmpty	$O(1)$	$O(1)$
Delete Entire Stack	$O(1)$	$O(1)$



Stack using Linked List

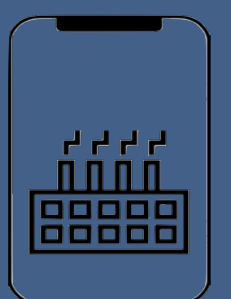
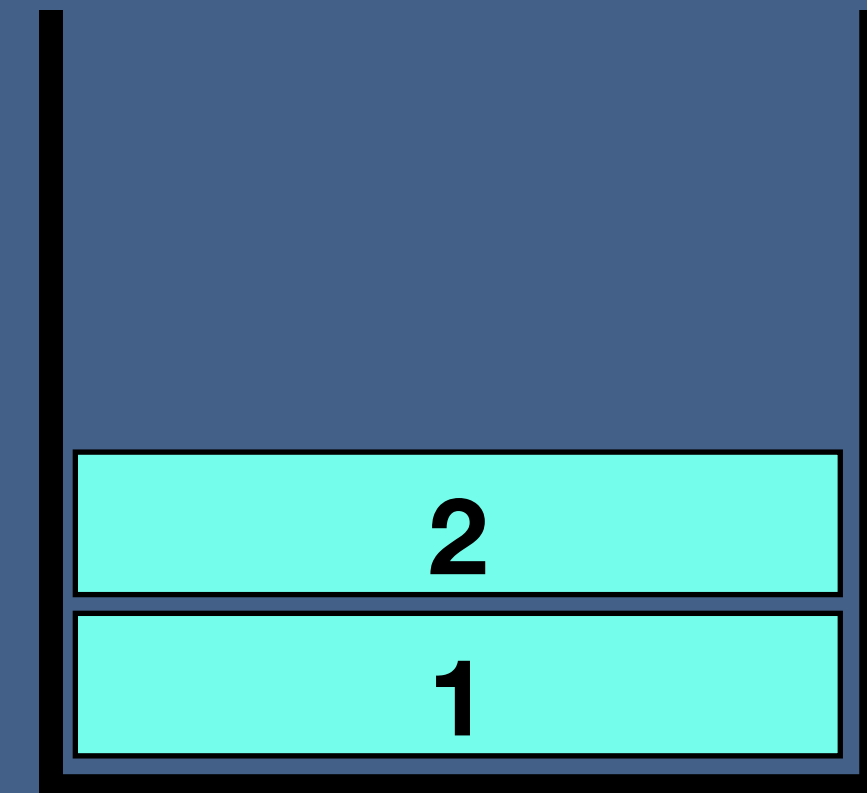
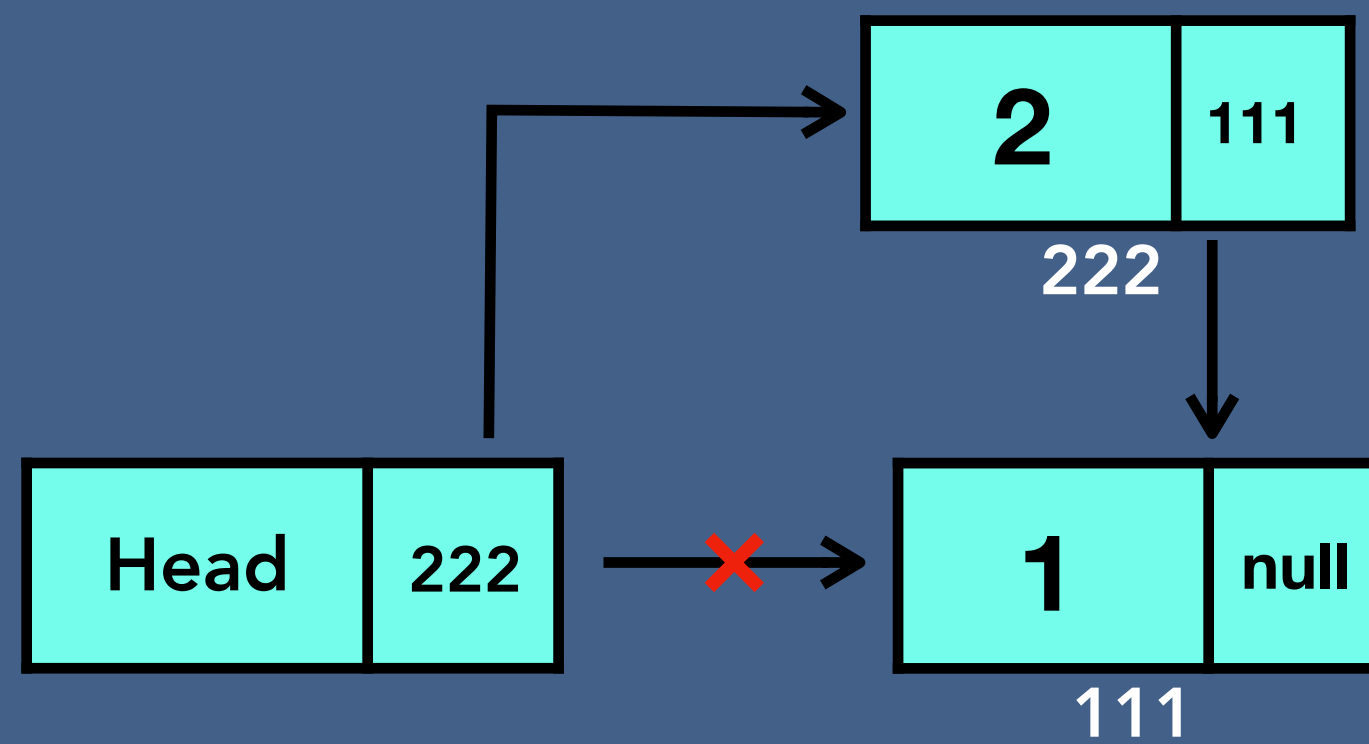
Create a Stack

Create an object of Linked List class



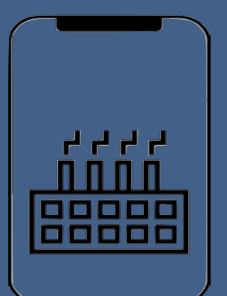
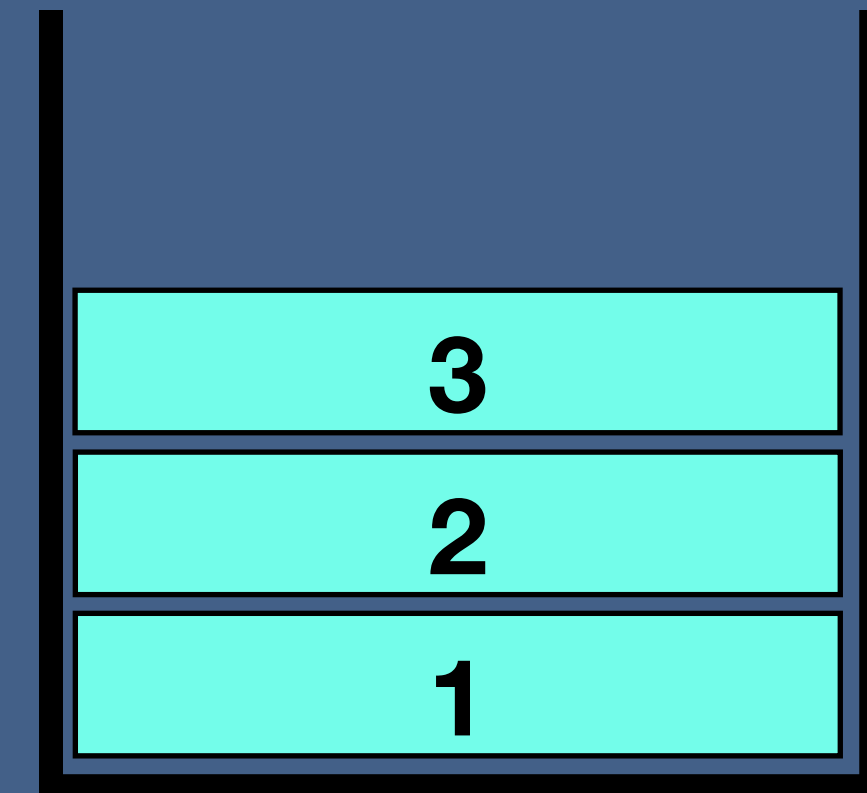
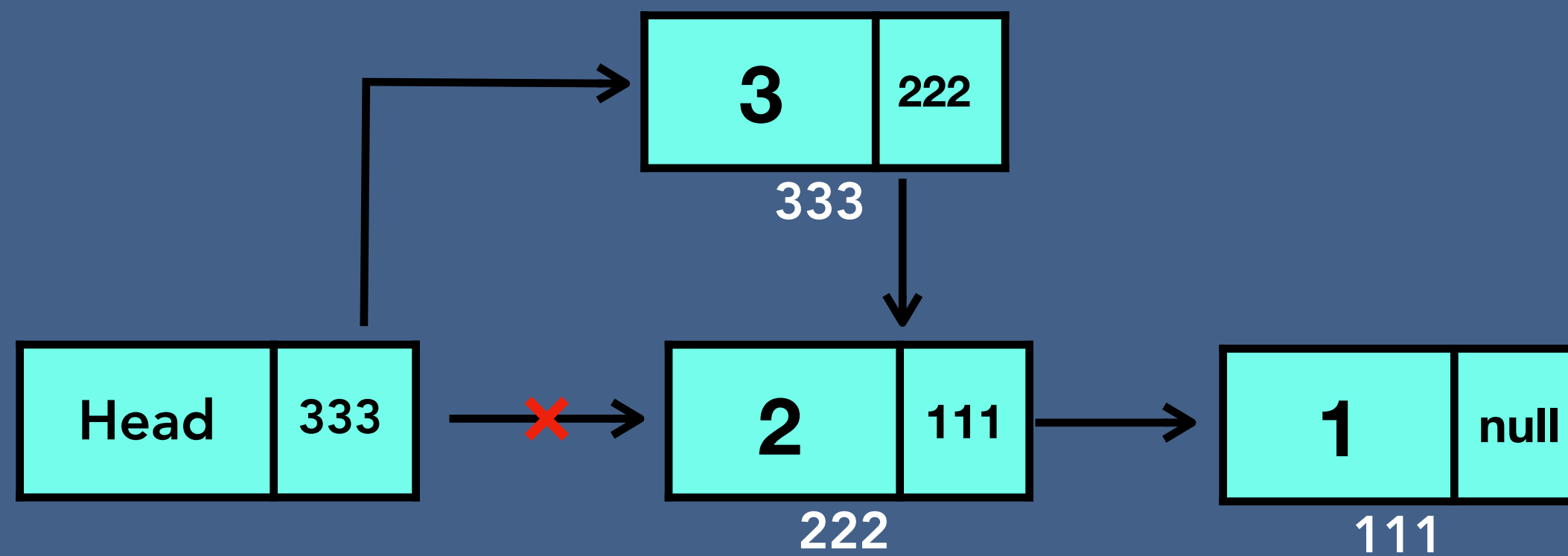
Stack using Linked List

Push() Method



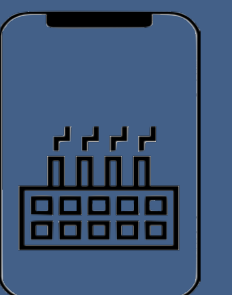
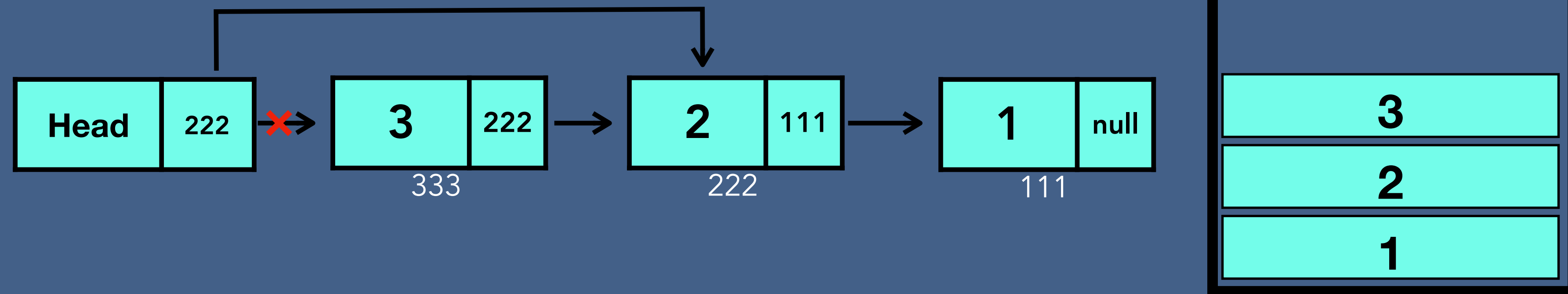
Stack using Linked List

Push() Method



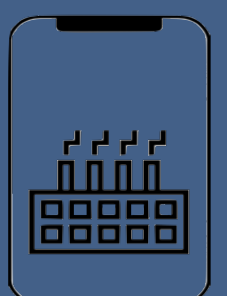
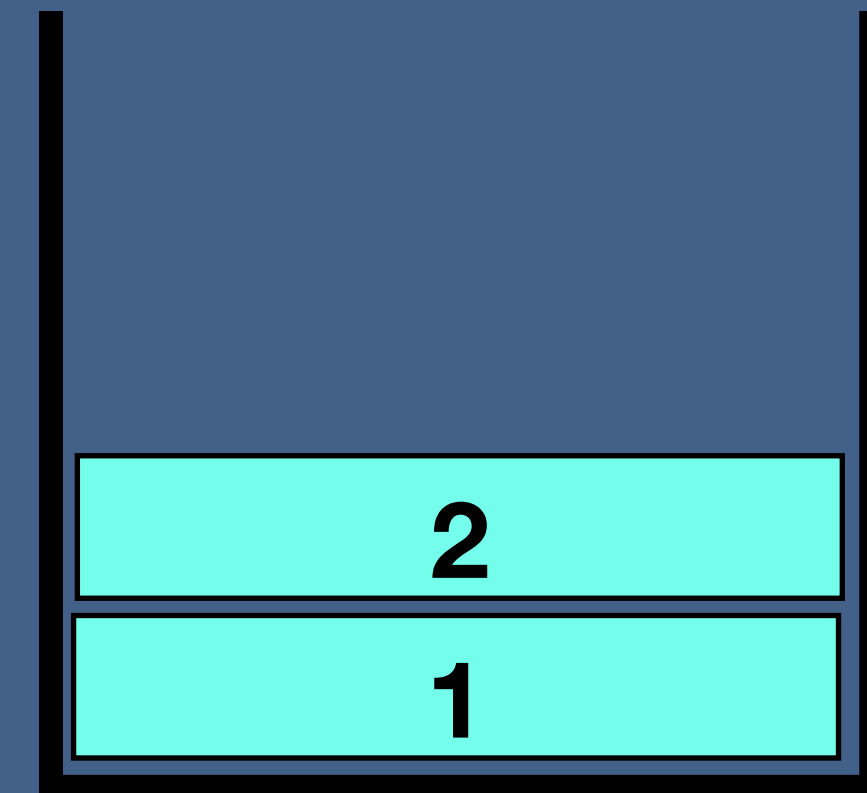
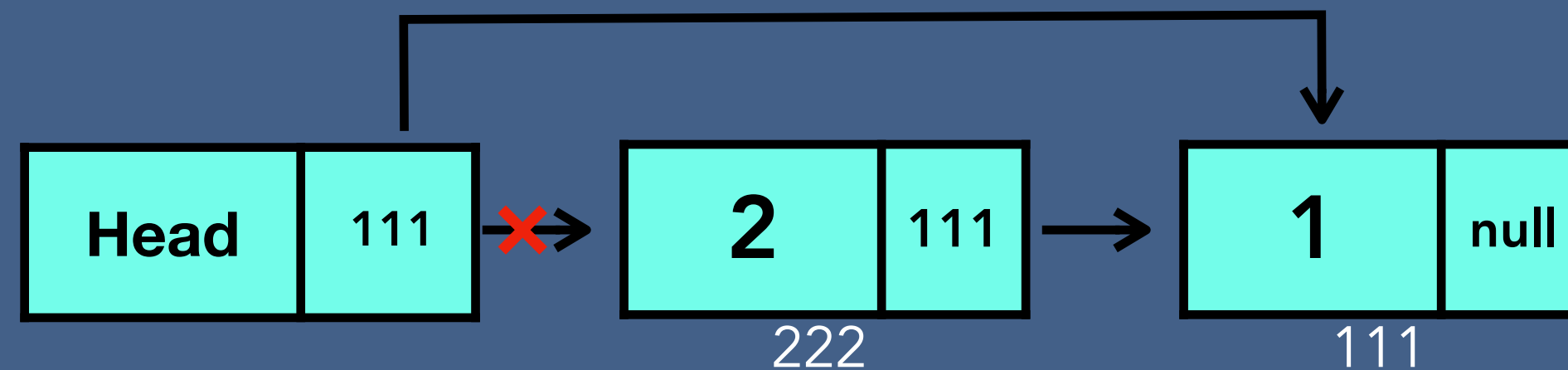
Stack using Linked List

Pop() Method



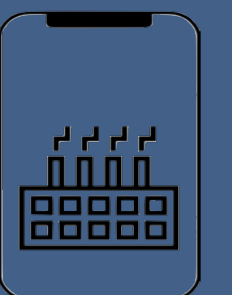
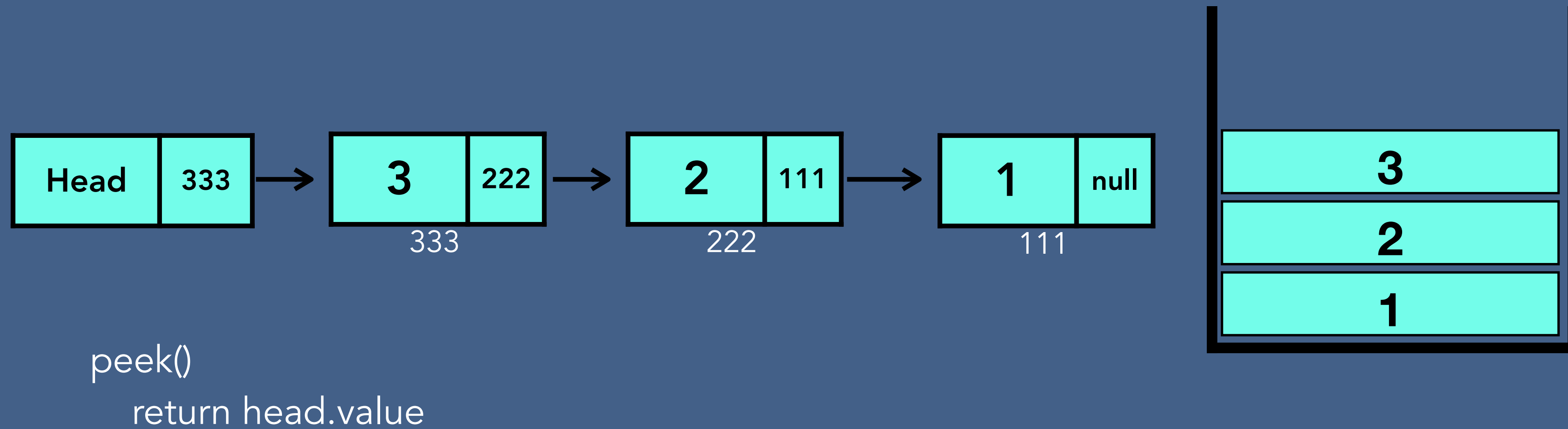
Stack using Linked List

Pop() Method



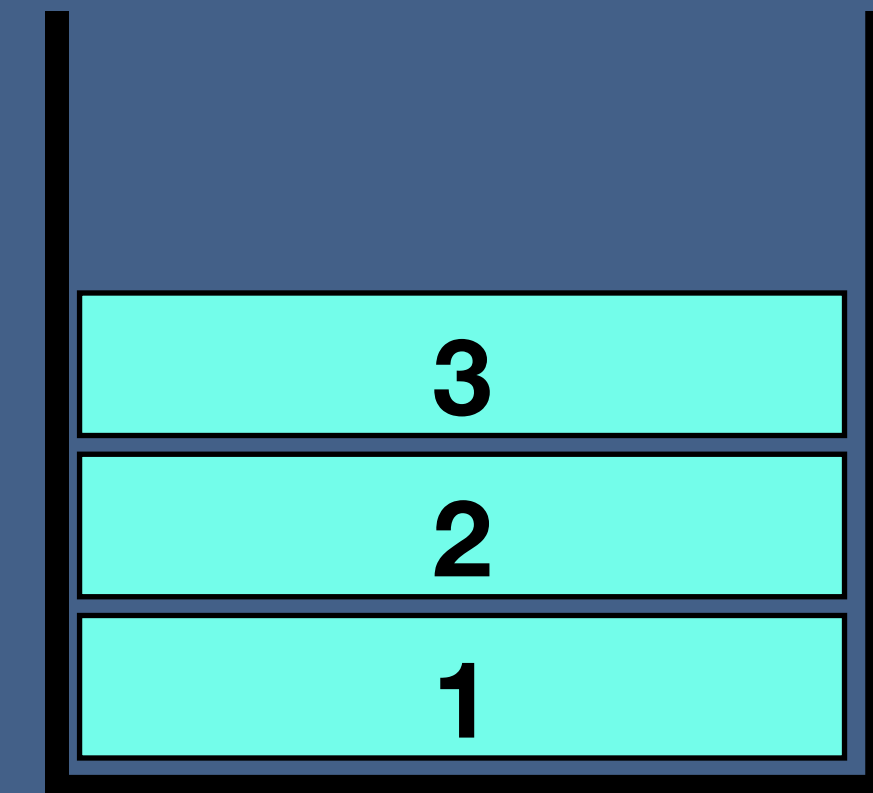
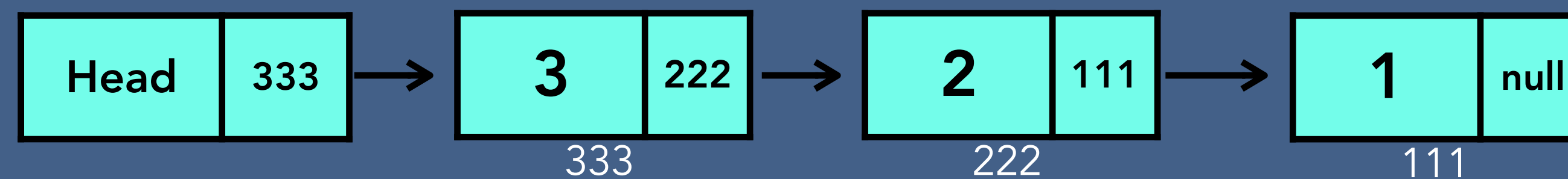
Stack using Linked List

Peek() Method

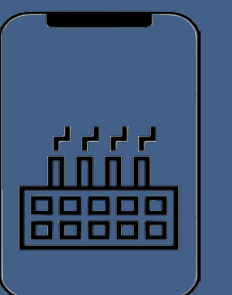


Stack using Linked List

isEmpty() Method

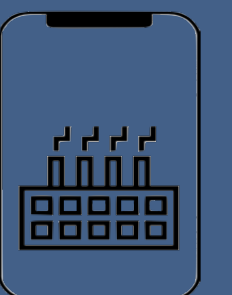
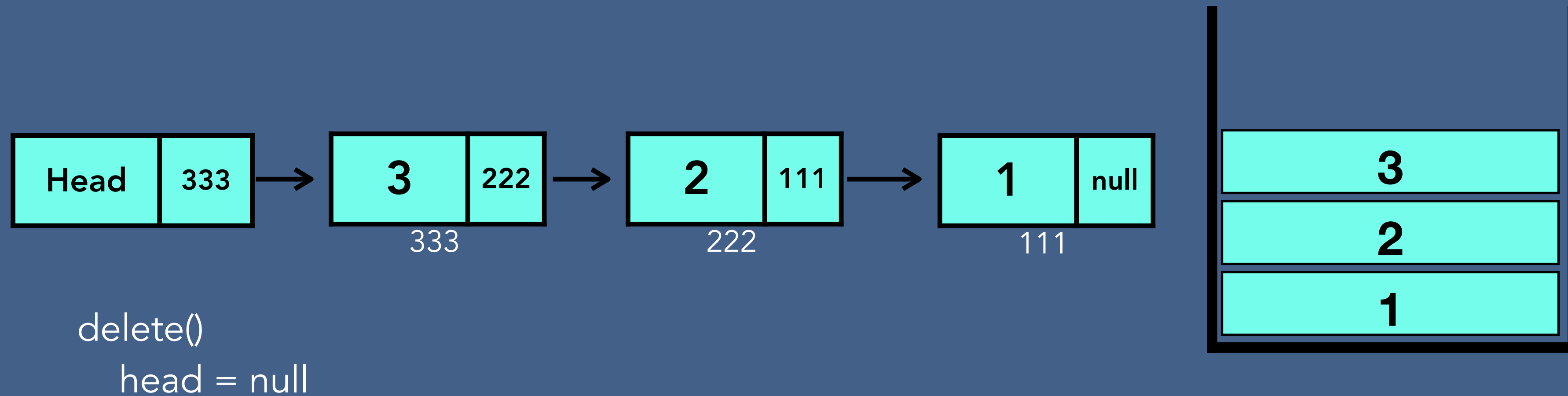


```
isEmpty()
{
    If head==null {
        true
    }
}
```



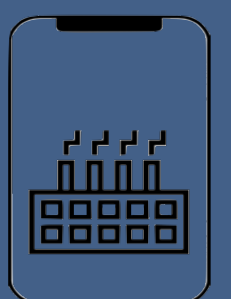
Stack using Linked List

delete() Method



Time and Space Complexity of Stack using Linked List

Stack	Time complexity	Space complexity
Create Stack	$O(1)$	$O(1)$
Push	$O(1)$	$O(1)$
Pop	$O(1)$	$O(1)$
Peek	$O(1)$	$O(1)$
isEmpty	$O(1)$	$O(1)$
Delete Entire Stack	$O(1)$	$O(1)$



When to Use/Avoid Stack

Use:

- LIFO functionality
- The chance of data corruption is minimum

Avoid:

- Random access is not possible

